

Case 03: Design and Development of an Interbody Spinal Cage Using SolidWorks

1. Introduction

An interbody spinal cage is an implant used in spinal fusion surgeries to restore disc height, maintain spinal alignment, and promote bone fusion between adjacent vertebrae. These cages provide immediate mechanical support while facilitating long-term biological fixation through bone growth.

2. Need for Spinal Cage

- Restoration of intervertebral disc height
- Stabilization of the spinal segment
- Decompression of nerve roots
- Maintenance of spinal alignment
- Promotion of bone fusion between vertebrae
- Reduction of back pain and spinal instability

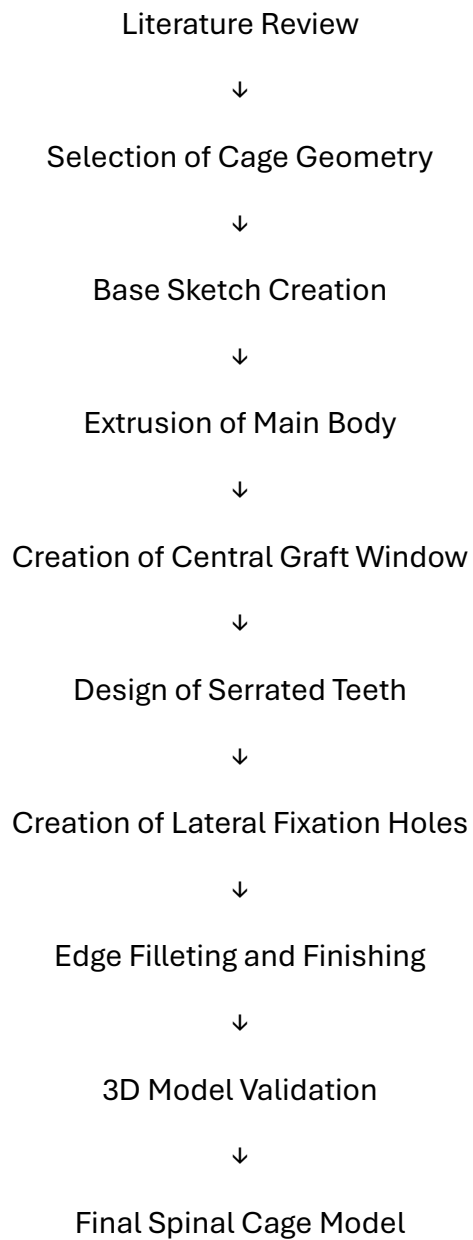
3. Objective

- To design a spinal interbody fusion cage using SolidWorks.
- To understand the geometry and functional features of spinal implants.
- To develop a model suitable for future additive manufacturing applications.
- To study implant design considerations for spinal fusion surgery.

4. Software Used

Component	Software
CAD Modeling	SolidWorks
Design Validation	SolidWorks
STL Generation	SolidWorks

5. Methodology



6. Design Features

Feature	Function
Hollow Central Region	Bone graft placement
Serrated Teeth	Prevent implant migration
Lateral Holes	Weight reduction and bone integration
Curved Surface	Anatomical conformity
Tapered Geometry	Easier insertion during surgery

7. Materials Used in Spinal Cage

Component	Material	Reason for Selection
Spinal Cage	Medical Grade PEEK	Elastic modulus similar to bone, radiolucent, biocompatible
Alternative Material	Ti-6Al-4V ELI	High strength and excellent osseointegration
Bone Graft Filling	Autograft / Allograft	Promotes spinal fusion
Fixation Screws (if used)	Titanium Alloy	Corrosion resistance and biocompatibility

8. Additive Manufacturing Compatibility

Parameter	Specification
Implant Type	Interbody Spinal Cage
Material	Ti-6Al-4V ELI / Medical Grade PEEK
Manufacturing Method	SLM (Titanium), FDM/SLS (PEEK)
Design Software	SolidWorks
File Format	STL
Post Processing	Surface Finishing and Sterilization

